



Randomized controlled trial of the early adolescent coping power program: Effects on emotional and behavioral problems in middle schoolers

Catherine P. Bradshaw^{a,*}, Heather McDaniel^b, Elise T. Pas^c, Katrina J. Debnam^a,
Jessika H. Bottiani^a, Nicole Powell^d, Nicholas S. Ialongo^c, Antonio Morgan-Lopez^e,
John E. Lochman^d

^a University of Virginia, Bavaro Hall, 112-D, 417 Emmet Street South, P.O. Box 400260, Charlottesville, VA 22904, USA

^b University of South Carolina, Department of Psychology, 1512 Pendleton St, Columbia, SC 29208, USA

^c Johns Hopkins Bloomberg School of Public Health, 615 N. Wolfe Street, Baltimore, MD 21205, USA

^d University of Alabama, Box 870348, The University of Alabama, Tuscaloosa, AL 35487, USA

^e Research Triangle Institute, RTI International, 3040 E. Cornwallis Rd., Hill 350PO Box 12194Research Triangle Park, NC 27709, USA

ARTICLE INFO

Editor: Andy Garbacz
Action Editor: Daniel B. Hajovsky

Keywords:
Prevention
Aggressive behavior
Randomized controlled trial
Middle school
Indicated programming

ABSTRACT

We report findings from a 40 middle school randomized controlled trial of an adapted version of Coping Power (Lochman & Wells, 2002a) for middle schoolers, called the Early Adolescent Coping Power (EACP) Program (Bradshaw et al., 2019) to determine the impact of EACP on adolescents' mental health outcomes, as indicated by student self-reported and teacher-ratings on the Behavior Assessment System for Children (BASC). The EACP was implemented over the course of students' 7th grade school year. The sample included 709 students who were identified at baseline through a teacher screening process for aggressive behavior and enrolled into the project (69.8 % African American and 59.4 % male). For teacher-reported outcomes, intent to treat (ITT) results indicated that EACP was associated with a decrease in externalizing problems over time. Exploration of effect modification suggested that girls in the EACP condition demonstrated statistically significant decreases in school problems over time, as well as a baseline by intervention effect whereby students and schools with greater baseline difficulties demonstrated decreased internalizing problems over time. For student-reported outcomes, there was a significant moderated effect, whereby girls in EACP demonstrated more favorable self-reported personal adjustment outcomes. Together, these results suggest that the early adolescent adaptation of Coping Power had preventive effects for 7th grade participants, and promotive effects specifically for girls, over time.

Aggressive and disruptive behavior problems often co-occur with poor academic functioning and low school connectedness (Herman et al., 2022; Reaves et al., 2018). These issues are particularly salient in middle school, where rates of school violence and disruption are higher relative to elementary school (Monachino et al., 2021; NCES, 2022) and student engagement and caregiver involvement in school-based programming decline (Wang & Eccles, 2012). Yet there are fewer school-based prevention programs to

* Corresponding author.

E-mail address: cpb8g@virginia.edu (C.P. Bradshaw).

address the behavioral and mental health concerns of middle school students, especially programs for students engaging in high rates of aggressive behaviors and demonstrating other indicators of risk for disengagement from school, academic problems, and longer-term mental health concerns (Evans et al., 2007; Evans et al., 2017; Farmer et al., 2015, also see Alperin et al., 2023; National Academies of Sciences, Engineering, and Medicine, 2019). Given the growing mental health crisis (Office of the Surgeon General, 2021; Weist et al., 2024), and schools' access to youth, it is a valuable context for delivering interventions to meet their behavioral and mental health needs (Evans et al., 2023; Santiago et al., 2018). Delivering preventive interventions in schools can increase access to care more generally, and address equity issues in the prevention of mental health problems among youth (see Alperin et al., 2023; Evans et al., 2007; Evans et al., 2023).

Coping Power (Lochman & Wells, 2002a) is a rigorously tested, school-based program that aims to prevent aggressive behavior problems in upper elementary students. This multi-component program includes clinician-facilitated group sessions for youth, separate group sessions for caregivers (i.e., parents), and select program activities for teachers. Though found to be effective with students in the upper elementary grades, prior to the present study, Coping Power had rarely been tested with youth in the middle school years, beyond 6th grade (for a review, see Lochman & Wells, 2002b; Lochman et al., 2021; McDaniel et al., 2023). Given that previous studies have demonstrated a range of positive effects for Coping Power with upper elementary school children (i.e., grades 4–6, within the elementary school setting), there was potential that these effects could also generalize to students in middle school. However, to increase the developmental fit of select program elements and activities, we made systematic developmental adaptations to the intervention, in turn creating a slightly adapted version of the model called the *Early Adolescent Coping Power* (EACP) Program (Bradshaw et al., 2017). Like the original, this adapted version of Coping Power was intended to meet the social, cognitive, and developmental needs of middle school students (grades 6–8) showing signs of difficulty self-regulating anger through supports to students, their caregivers, and teachers, to prevent further escalation of conduct and related behavior problems. Like earlier versions of Coping Power, EACP was designed to be delivered by school-based clinicians and counselors through schools during the school day, to overcome traditional barriers to services and inequities in access to services, including for students of color. EACP aims to prevent the escalation of these concerning behavior problems and avoid out of school placement for more intensive services (Lochman & Wells, 2002a; National Academies of Sciences, Engineering, & Medicine, 2019).

The current study reports findings from a school-level randomized controlled efficacy study of EACP with students in 7th grade to determine impacts on teacher-rated externalizing problems at post-intervention and at a one-year follow-up. Secondary outcomes of interest included possible impacts on student self-reported concerns and the potential for cross-over outcomes to internalizing symptoms, as such effects have rarely been examined in prior studies. We also explored select effect modifiers beyond the intent to treat effects, including possible variation by race, sex, and baseline scores.

1. Background on coping power

Coping Power was designed to be an “indicated” prevention program (National Academies of Sciences, Engineering, & Medicine, 2019), to support students displaying aggressive behavior problems, with the goal of preventing later, and possibly worse, school and behavioral problems. Our focus in the current study on middle school-aged students is especially important, given the developmental literature highlighting the significance of this age group for concerns related to bullying, truancy, dropout, school failure, and trajectories of aggressive behavior problems (e.g., Farmer et al., 2015; Farrell et al., 2018; Steinberg, 2016). Psychological functioning, self-esteem, and academic performance often tend to decline in the year following the middle school transition (Eccles et al., 2018). Youth also display increases in substance use, delinquency, aggression, and other problem behaviors in middle school (Farrell et al., 2018; NCES, 2022; Steinberg, 2016).

Originally designed for children in upper elementary school (grades 4–5), Coping Power is based on a contextual social-cognitive developmental model (Lochman & Wells, 2002a). The multi-component program targets youth social cognitive and self-regulation processes which are hypothesized to mediate impacts on student outcomes. Similarly, a set of program activities occur for family/caregivers (referred to in the current paper collectively as caregivers) which focus on practices and processes that aim to improve caregiver/child relationships and behaviors in the home environment, which extend into the community. A third set of activities target select teachers and focus on improving classroom-based supports. Specifically, these teachers engage in a series of activities to promote school bonding and academic organizational skills, with the goal of promoting academic success. For additional review of the Coping Power theoretical framework and the conceptualization of the mechanisms of change, see Lochman and Wells (2002a) and Lochman et al. (2008). In the sections below, we describe the various components of the EACP version of Coping Power tested in the current study. We highlight some of the systematic developmental adaptations made to this version of Coping Power for youth in middle school (i.e., grades 6–8; see Bradshaw et al., 2017).

2. Overview of the EACP components tested in the current RCT

The *Early Adolescent Coping Power* (EACP; Bradshaw et al., 2017) version, like the original, is comprised of a student, caregiver, and teacher component and is based on a contextual social-cognitive developmental model (Lochman & Wells, 2002a). It was systematically adapted and feasibility tested through a NIMH funded grant to: (a) include fewer student sessions (i.e., reduced from 34 to 25), (b) be delivered over a single school year, and (c) incorporate content applicable to early adolescent students in grades 6 through 8. In the sections below, we describe the specific components and briefly summarize some of the adaptations made in creating the EACP version (for additional details on the adaptations see Bradshaw et al., 2017). For an overview of the original Coping Power youth curriculum and the theorized mechanisms of change see Lochman et al. (2008).

The **EACP Youth Component** as delivered in the context of this randomized controlled trial (RCT) included both a group and individual component. The youth group component included 6–7 students and took place at the students' schools, typically during nonacademic homeroom periods, recess, specials, or lunch. The 25 group sessions lasted approximately 45 min and were designed to fit within a single course period. Group youth sessions delivered in this RCT were held approximately once week over the school year (spanning nearly 9 months). The group sessions in this RCT were co-led by a grant-funded intervention staff member (i.e., master's, doctorate-level clinician, or clinician trainee) and when possible, were co-facilitated by the schools' counselor, social worker, or school psychologist. Together, the group sessions were designed to foster students' social-cognitive development (Lochman et al., 2008).

The first eight sessions (Phase 1) focused on the delivery of content and concepts to the students and on developing group norms, setting short- and long-term goals, organizational and study skills, awareness of feelings, anger coping and self-control, coping self-statements, relaxation techniques, and perspective-taking. In the 9th through 15th session (Phase 2), students learned to apply the concepts taught by practicing perspective-taking and social problem-solving using the "PICC" (Problem Identification Choices Consequences) model. This phase culminated in the student group planning and creating a video to teach others the concepts that they had learned. In the final phase (i.e., sessions 16–25), developmental adaptations were made to support the students in applying their skills to scenarios and challenges early adolescents face, including friendship and romantic relationship development, social aggression, cyberbullying, problem-solving about damaged relationships and neighborhood problems, refusal skills/coping with peer pressure, deviant peer groups and centrality of group membership, positive quality development, entering new peer groups, and using positive peer networks. Students were also taught to utilize assertive "I-statements."

In addition, EACP was designed so that each child also received a total of 8–10 individual 30-min sessions with the group leaders during school hours to further build rapport, monitor progress, and reinforce content (see Lochman et al., 2008). The individual sessions were led by the clinician to encourage generalization of the skills developed during the group sessions (see Lochman et al., 2008). The individual sessions were used primarily for monitoring and reinforcing attainment of classroom and social behavior goals (e.g., avoiding fights, resisting peer pressure) and coping with specific attributional biases and social problem-solving challenges.

The **EACP Caregiver Component** consisted of 12 group sessions for caregivers, implemented over the same school year as the student component. These sessions focused on building caregiver skills in behavior management, supporting the new skills students were learning in student groups, and to promote a more positive school experience for students. Caregivers met in groups of 10–12 with the two co-leaders of the youth component. The EACP caregiver component was derived from social learning theory-based caregiver training programs developed and evaluated by prominent clinician-researchers in the field of child aggression (Lochman et al., 2008). Over the course of the 12 sessions, caregivers received training on a variety of skills including rewarding wanted behaviors, giving effective instructions, establishing age-appropriate rules and expectations for adolescents, applying effective consequences to unwanted behavior, establishing ongoing family communication structures (e.g., weekly family meetings), and effective monitoring. Efforts were made to include all caregivers in groups; we incorporated multiple engagement activities, such as motivational interviewing strategies (Herman et al., 2012), to increase engagement and caregiver attendance. Additional structural modifications and activities were provided to increase attendance, such as offering caregiver sessions at flexible times (e.g., after school and in the evening). We also offered transportation / taxi vouchers, child friendly activities for siblings during the sessions, and food to increase parental attendance.

In addition to these "standard" caregiver skills, caregivers also learned to support the social-cognitive and problem-solving skills that youth learned in the EACP youth component. Caregiver skills were introduced at the same time as the respective youth skills, so that caregivers and youth could work together at home to reinforce what they were learning in the program. For example, caregivers learned to set up homework support structures and reinforce organizational skills. Caregivers learned to support students' bonding to school and methods for positive involvement in the school setting (e.g., setting up productive, proactive meetings with teachers, and maintaining contact). Caregivers also learned to apply the social problem-solving model to the family so that the students' skills learned in the group would be prompted and reinforced at home. The sessions also addressed parenting adolescents, communication, and some issues related to bullying and peer relationships among adolescents.

Finally, the **EACP Teacher Component** targeted identified "cheer teachers" for the Coping Power intervention to support their generalization of skills and to monitor progress. A teacher was identified for each participating student (i.e., in some cases, one teacher served as a "cheer teacher" for more than one student; on average teachers served as the cheer teacher for 1–4 students, with a modal number of 3 students per teacher). The teachers were selected if they were core content teachers (e.g., language arts, mathematics) or homeroom teachers, to ensure they would have sufficient access to the participating students. These teachers engaged in three 30-min meetings during the school year with an EACP clinician. These meetings included a didactic presentation on EACP, reviewing its theoretical approach, the structure of student sessions, and identification of the key skills students were learning so that teachers could support and optimize their skill use in the classroom. Teachers received instruction on how to support students in setting goals, reviewing weekly tip sheets of topics and activities, discussing goal progress and challenges, encouraging students to apply skills developed in the program, building positive relationship with students, and communicating with the school counselor and EACP clinician about student concerns and progress. Teachers received written instructions on how to implement these activities; the implementation of the EACP teacher content was manualized. Cheer teachers were expected to deliver EACP teacher content during their routine interactions in the classroom to help promote generalization in the classroom setting and within the peer group.

3. Prior research on coping power

Although this is the first efficacy study of EACP, several RCTs of the original version of Coping Power have demonstrated significant improvements in younger children's disruptive behavior problems, both in the short and long term (e.g., 4-year follow-up) (e.g.,

Lochman et al., 2013; Lochman & Wells, 2002b, 2003, 2004). These earlier studies with younger children have also shown preventive effects on associated outcomes, such as reduced substance use (Lochman & Wells, 2003, 2004), and improved grades (Lochman et al., 2012). Specifically, Coping Power reduced aggression at the end of the intervention period and at 1-, 3-, and 4-year follow-ups, and improved school behavior outcomes (e.g., reductions in disciplinary referrals, suspensions) and student concentration, on-task behavior, and study skills (Lochman et al., 2013; Lochman & Wells, 2002b, 2003, 2004). The efficacy studies of the full-length, multicomponent Coping Power program in comparison to school-as-usual comparison groups at one-year follow-ups after intervention (Lochman & Wells, 2003, 2004) found small effect sizes for intervention effects on covert delinquency (effect sizes [ES] = 0.28–0.36) and on teacher-rated behavior problems (ES = 0.34–0.35), and medium effect sizes for substance use (ES = 0.58–0.64). Placing these findings in context, school-based prevention studies that have effectively reduced crime, substance use and conduct problems typically have small effect sizes (Wilson et al., 2001).

A selective meta-analysis by McDaniel et al. (2023) examining a set of 12 pre-post studies of adapted versions of Coping Power across a broad range of outcomes (internalizing and information-processing, in addition to the primary externalizing outcomes) and applications (including when delivered as a universal intervention, as a briefer intervention, as a dissemination study, or within a specialized school for deaf children) of Coping Power demonstrated “small to moderate reductions in externalizing behavior, although these findings were not statistically significant” (McDaniel et al., 2023, p. 350); specifically, they noted an “overall mean effect size of $g = -.15$ ($p = 0.252$; $CI[-0.42, 0.12]$ ” (McDaniel et al., 2023, p. 365) based on 55 effect sizes based across 12 studies. The lack of statistical significance despite meaningful effect sizes may be due to limited power for statistical testing in meta-analyses with relatively few studies (Pigott & Polanin, 2015).

Prior pilot research on EACP. With funding through a NIMH grant, the EACP was feasibility and acceptability tested, and pilot tested in a subsequent under-powered study with 30 middle school students (Bradshaw et al., 2017). Pre-post-intervention results on the teacher-report version of the Behavioral Assessment for Children- 2nd Edition (BASC-2) indicated that the program was associated with reductions in externalizing problems and hyperactivity, and improvements in adaptive skills and leadership (Bradshaw et al., 2017). Together, the findings from previous RCTs of the original Coping Power program and the pilot work on EACP provided the rationale for testing the EACP with middle schoolers in a more rigorous study examining a broader set of outcomes.

4. Overview of the current study

The current study presents findings from a school-level RCT of EACP as implemented in seventh grade. We report findings contrasting baseline of 7th grade with the end of 7th grade, and at one-year follow-up (end of 8th grade) on students' behavior at school. We assessed outcomes based on student self-report and teacher report to determine the impact of EACP on social, emotional, and behavior problems. Most Coping Power research to date has focused on teacher ratings of student behavioral outcomes, but little research has included student self-reported behaviors, likely due to the historical focus on younger student participants in the Coping Power program. Our primary aim focused on intent to treat analyses to determine the main effects of the EACP preventive intervention on key teacher-reported externalizing outcomes of interest that put youth at risk for a range of behavioral and mental health problems. Consistent with prior work by Lochman and Wells (2002b, 2003, 2004) (also see Lochman et al., 2013; McDaniel et al., 2023), we anticipated that EACP would have a significant impact on teacher ratings of externalizing problems, which was our primary outcome of interest.

We were also interested in exploring for the potential of youth self-reported outcomes of externalizing problems, and the possibility of cross-over effects on internalizing symptoms (Clifford et al., 2020). The inclusion of youth self-report data and internalizing problems are an important and unique contribution of the current study. Specifically, this assessment of internalizing symptoms allowed for the exploration for transdiagnostic effects of EACP on depression and anxiety, as there is some research suggesting that Coping Power may have an impact on these factors in addition to the typical impacts on externalizing outcomes (Clifford et al., 2020). Documenting such impacts may suggest a broader set of intervention effects on mental health outcomes for EACP than previously assessed, which is especially timely given the rising rates of mental health concerns among youth (Office of the Surgeon General (OSG), 2021; Weist et al., 2024). Given these effects were exploratory and secondary, we did not have specific hypotheses about these outcomes. Finally, we explored how the impact of EACP may have been moderated by sex, race, and baseline levels of problem behaviors.

5. Method

5.1. Study design

We conducted a school-level group RCT (Murray, 1998) to evaluate the effects of EACP on 7th grade students' behavior in 40 middle schools in Alabama and Maryland, with 20 schools enrolled in each state. The total sample of 40 schools was recruited into the project through district partnerships with the researchers. Half of the schools in each state were randomized to implement the intervention and half were assigned to school as usual, just once, prior to the screening of the first year of the project. Schools were randomized to either a control or intervention condition to avoid contamination effects that could occur if students were randomized within schools. We created 20 matched pairs of schools and randomly assigned schools within pairs to EACP and control. Participating schools had a mean enrollment of 569.03 students ($SD = 238.11$) and racially diverse student populations (e.g., $M = 58.02$ % African American, $SD = 34.44$). In addition, schools reported, on average, that 68.06 % ($SD = 26.33$) of students in their school were eligible for free or reduced-price meals.

Three consecutive cohorts of approximately 120 students each year were included in the intervention condition and contrasted to

the 120 control students each year. Data were collected from intervention and control students at three time points over the course of two years (one intervention year and one follow-up year). The students identified for inclusion in the study were selected from the approximately 25 % of rising seventh graders in the sample who evidenced the highest levels of aggression relative to their peers, according to teacher ratings (see details on screening procedure below). Identical screening procedures were used in both conditions; however, in the intervention schools, those youth who were screened and consented to participate entered an EACP intervention group (comprised of 6–7 students). One group was formed in each intervention school and the intervention was provided to the students over the course of a school year. This same process was conducted with a new cohort of 7th grade students in cohorts 2 and 3. The data for this project were collected between fall of 2014 and spring of 2018 (follow up).

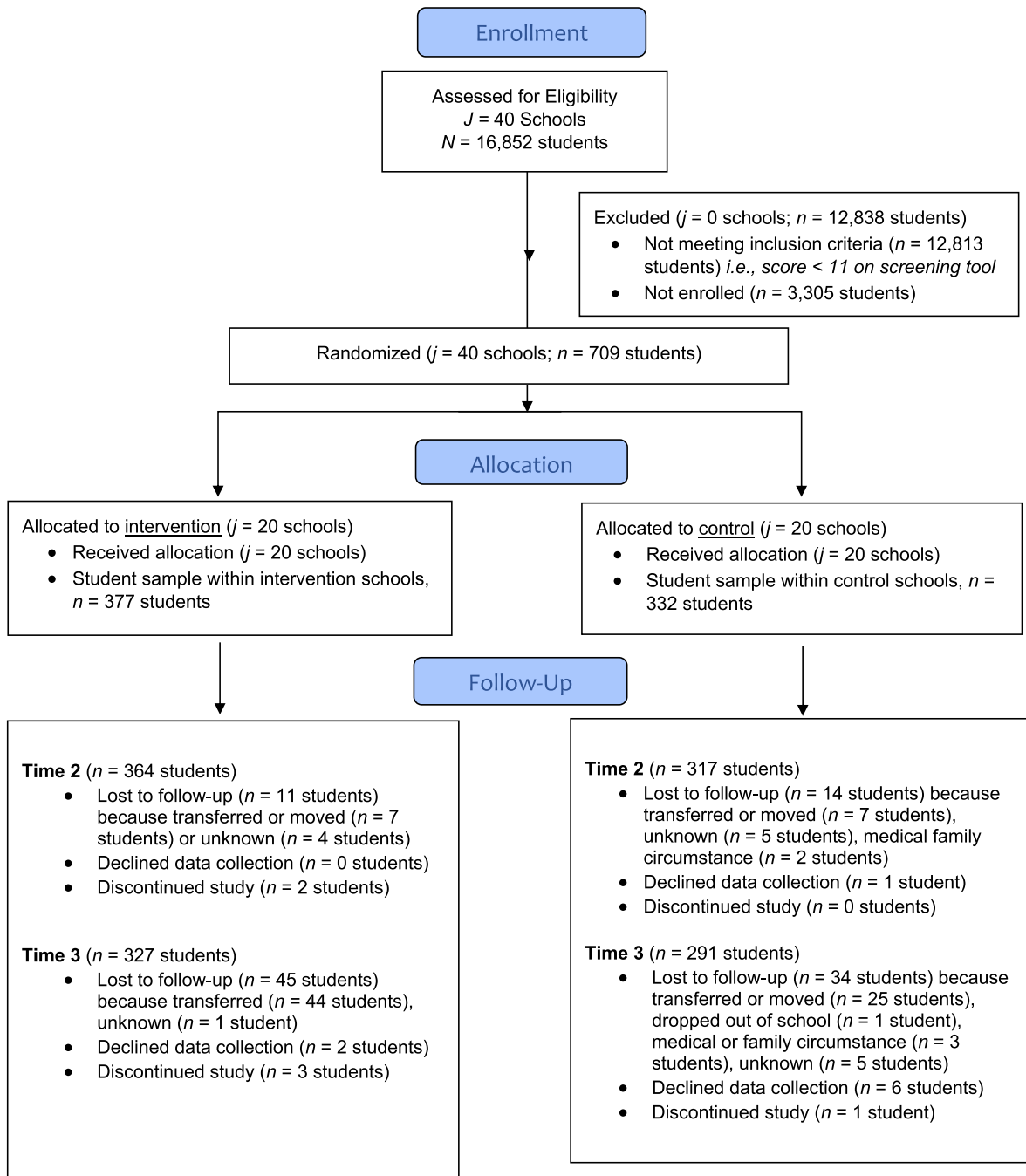


Fig. 1. CONSORT diagram for the Middle School Coping Power School Cluster Randomized Trial.

5.2. Participants

The sample included 709 7th graders. In the 20 schools randomized to implement the program, 10 in Alabama and 10 in Maryland, 377 youth were allocated to receive the intervention whereas 332 students were allocated to participate in school-as-usual (see CONSORT diagram in Fig. 1). The final sample of participating students was 69.82 % African American and 59.38 % male. At baseline, over 50 % of the sample was in the at-risk or clinically significant range for low adaptive skills and high externalizing problems as measured on the Behavioral Assessment for Children, 2nd Edition (BASC-2 Adolescent, ages 12–21; Reynolds & Kamphaus, 2004); approximately 20 % was at-risk for or clinically significant for internalizing problems on the BASC-2. Additional demographic information, including baseline levels of emotional and behavioral difficulties, is reported in Tables 1 and 2.

5.3. Procedures

5.3.1. Screening

During the previous spring academic semester, sixth grade teachers completed a universal, six-item screening measure (Reactive and Proactive Aggression Questionnaire [Dodge et al., 1997]) assessing rising 7th grade student proactive and reactive aggression (Lochman & Conduct Problems Prevention Research Group, 1995). Students who received a score of 11 on a possible range of 6–30 on proactive and reactive aggression (i.e., tending to be ranked in the highest quarter [25 %] of students) were recruited to begin participation in the study during the following year (i.e., the fall of seventh grade) and comprised the eligible study sample. Participation was voluntary. A list of eligible students was generated based on their screening data and parents were contacted by project staff using the contact information provided by the school. Children were enrolled on a first come first serve basis until the seven EACP group positions were filled per school, first obtaining the parental consent for their child's participation and for their own (parental) participation, and then the youth assent. After the participants were enrolled in the program, the baseline data were collected from the teachers, youth, and parents (as described below). All study data were collected in fall of 7th grade at baseline prior to beginning EACP intervention groups, at the end of the 7th grade school year (post-test), and in the spring of 8th grade as the one-year follow-up. All measures are described below. The study was approved by the institutions' Institutional Review Boards.

5.3.2. EACP implementation fidelity data

The intervention groups were conducted during the students' 7th grade year. On average, students attended 78.82 % of sessions (i.e., a mean of 18.58 out of the 25 sessions; valid range 0 to 25; $SD = 6.07$). Student attendance data showed that 57.29 % of the students attended 80 % or more of the sessions and 81.7 % of the students attended 60 % or more of the sessions. Approximately 61.7 % of the groups received all 25 sessions; an additional 16.7 % of the groups had 24 sessions delivered; 8.3 % had 23 sessions delivered, 3.3 % had 22 sessions delivered; and 10 % had 21 sessions delivered in this study. Students also received an average of 3.10 individual contacts with the clinician ($SD = 2.22$, range = 0–15), and spent an average of 65.65 total minutes with the group leader in these contacts ($SD = 70.75$, range = 0–510 min), indicating high adherence to the individual component of EACP. To assess youth participant engagement, group leaders answered 8 items from the Groupwork Engagement Measure (Macgowan, 1997) regarding each student in the group. This measure has been used in prior counseling research (e.g., Prado et al., 2006), with higher scores on this four-

Table 1
Demographic characteristics of students.

Student Characteristics (N = 709)	Control (N = 332) N (%)	Intervention (N = 377) N (%)
Sex*		
Male	188 (56.63 %)	233 (61.80 %)
Female	144 (43.37 %)	144 (38.20 %)
Race/Ethnicity*		
Native American/American Indian	5 (1.51 %)	8 (2.12 %)
Asian/Pacific Islander	2 (0.60 %)	2 (0.53 %)
Black/African American	224 (67.47 %)	271 (71.88 %)
White	71 (21.39 %)	63 (16.71 %)
Hispanic/Latino	10 (3.01 %)	15 (3.98 %)
Native Hawaiian or Other Pacific Islander	–	1 (0.27 %)
Other or more than one race	20 (6.02 %)	17 (4.51 %)
Free or reduced-price lunch		
No	19 (5.72 %)	31 (8.22 %)
Yes	170 (51.20 %)	186 (49.34 %)
I don't know	134 (40.36 %)	157 (41.64 %)
Missing	9 (2.71 %)	3 (0.79 %)
Cohort		
Cohort 1	121 (36.45 %)	126 (33.42 %)
Cohort 2	107 (32.23 %)	127 (33.69 %)
Cohort 3	104 (31.33 %)	124 (32.89 %)

Note. *These variables have no missing data because values were imputed from sources in the following order: as reported by the student in the student survey, then parent survey, then teacher survey, and finally from school records.

Table 2
School-aggregated baseline characteristics by intervention condition.

	Control ($J_{Schools} = 20$)		Intervention ($J_{Schools} = 20$)			
School Characteristic	M	SD	M	SD	t-value	p-value
<i>School demographic characteristics</i>						
Total enrollment (count)	504.45	208.52	633.60	253.22	1.76	0.086
African American (%)	62.27	34.87	53.77	34.37	−0.78	0.443
Free or reduced-priced meals (%)	73.85	25.84	63.36	26.41	−1.27	0.212
Average reading rate (%)	71.82	16.63	76.27	16.59	0.85	0.402
<i>Teacher-reported BASC-2 composites (T-scores)</i>						
Externalizing Problems	59.68	5.45	63.58	5.78	−2.20	0.034
Internalizing Problems	51.25	4.38	54.24	4.92	−2.03	0.049
Adaptive Skills	42.20	3.80	40.30	2.55	1.86	0.071
School Problems	57.45	4.05	59.65	3.56	−1.83	0.075
<i>Student-reported BASC-2 composites (T-scores)</i>						
School Problems	50.86	3.04	51.73	3.00	−0.91	0.369
Internalizing Problems	52.04	3.48	52.43	3.27	−0.36	0.722
Inattention/Hyperactivity	55.50	3.75	56.17	4.29	−0.53	0.598
Personal Adjustment	50.88	2.71	49.48	3.51	1.41	0.167

point scale indicating more engagement. The range was 1.38 to 4, with the overall mean of 3.06 ($SD = 0.41$). For additional details on these fidelity measures, see [Pas et al. \(2020\)](#).

There was high clinician adherence to the program, whereby clinicians implemented 24.13 (out of the possible 25) sessions on average and implemented 71.81 % of all objectives for all 25 sessions. It is worth noting there are extra activities and objectives in each session to allow for some clinician discretion. For additional details on EACP implementation, see [Pas et al. \(2020\)](#) (also see a prior developmental study by [Bradshaw et al., 2017](#) regarding program feasibility and acceptability). Caregivers' attendance ranged from zero to all 12 full sessions, with 40.11 % of caregivers attending one or more sessions. On average, caregivers attended 2.08 of the 12 sessions.

5.4. Equivalence at baseline

Given the school randomized design, we examined baseline equivalence of intervention and control schools at the school level. Specifically, we examined between-group baseline differences on school demographic characteristics, school-level student-reported BASC-2 composites, and teacher-reported BASC-2 composites (see means, standard deviations, and results of the *t*-tests in [Table 2](#)). Results of the school level independent samples *t*-tests suggested that there were no statistically significant differences on school demographics, or student-reported BASC-2 composites. There were, however, significant baseline differences on two of the teacher-reported BASC-2 scores (i.e., Externalizing Problems composite, $t(38) = -2.20$, $p = .034$, $d = 0.69$, and Internalizing Problems composite, $t(38) = -2.03$, $p = .049$, $d = 0.64$), such that the intervention group schools demonstrated significantly more difficulties at baseline. However, the independent *t*-tests for the teacher-reported Adaptive Skills, $t(38) = 1.86$, $p = .071$, and School Problems, $t(38) = -1.83$, $p = .075$, were not statistically significant at the $p < .05$ level.

5.5. Measures

5.5.1. Student behavior

Teachers and students reported on students' emotional and behavioral health using the Behavioral Assessment System for Children (BASC-2: Adolescent, ages 12–21; [Reynolds & Kamphaus, 2004](#)). The BASC-2 has extensive research documenting reliability and validity evidence for the youth and teacher reports ([Reynolds & Kamphaus, 2004](#)). Both the student- and teacher-reported scales were collected at baseline, post-intervention, and at one-year follow-up. Teachers and students rated the frequency of each behavior on a four-point Likert scale ranging from *never* (0) to *always* (3).

Teacher-Report. We focused on the three clinical composites from the *Teacher Rating Scales*. The Externalizing Problems composite scale is comprised of items assessing hyperactivity, aggression, and conduct problems ($\alpha_{T1 - T3} = 0.969\text{--}0.971$). The Internalizing Problems composite is comprised of items assessing anxiety, depression, and somatization ($\alpha_{T1 - T3} = 0.891\text{--}0.911$). The School Problems composite is comprised of items assessing attention and learning problems ($\alpha_{T1 - T3} = 0.915\text{--}0.923$). In addition to the problem-related scales, teachers also report on adaptive skills. The Adaptive Skills composite includes items assessing adaptability, functional communication, leadership, social skills, and study skills ($\alpha_{T1 - T3} = 0.961\text{--}0.969$).

Student-Report. Similarly, the BASC-2 *Self-Report of Personality* reports three problem-related composites. The School Problems composite includes items assessing student attitudes towards school and teachers, as well as sensation seeking ($\alpha_{T1 - T3} = 0.752\text{--}0.754$). The Internalizing Problems composite assesses a variety of clinical problems including anxiety, depression, somatization, atypicality,

locus of control, social stress, and sense of inadequacy ($\alpha_{T1-T3} = 0.908\text{--}0.928$). The Inattention/Hyperactivity composite includes items related to attention and hyperactivity ($\alpha_{T1-T3} = 0.699\text{--}0.750$). The student-report survey also includes an assessment of the student's inter- and intra-personal strengths. The Personal Adjustment composite is comprised of items assessing relationships with caregivers, interpersonal relationships, self-esteem, and self-reliance ($\alpha_{T1-T3} = 0.825\text{--}0.861$). *T*-scores were calculated using the BASC scoring manual; composite scores were derived based on all scale scores. *T*-scores on the teacher- and student-reported problem-related composites greater than 60 indicate "at-risk" level difficulties and *t*-scores greater than 70 indicate clinically significant problems with impairment across multiple settings. *T*-scores on the adaptive composites of less than 40 suggest "at-risk" levels of adaptive skills and scores less than 30 are indicative of a clinically significant absence of adaptive skills.

5.5.2. Student demographics

Student demographic data (e.g., sex, race/ethnicity) were collected from students and/or teachers in conjunction with the data on the BASC-2.

5.6. Analytic plan

A series of bivariate tests of predictors of missingness/loss-to-follow-up were conducted for all four student-reported and four teacher-reported outcomes at post- and follow-up. For teacher-reported outcomes, intervention assignment, and site were significantly associated with missingness at post-assessment, while site, baseline BASC-2 scores, and student race were significantly associated with missingness at the one-year follow-up. For student-reported outcomes, intervention assignment, student race, and cohort were associated with missingness at post-assessment, while site and baseline BASC-2 scores were associated with missingness at follow-up. Covariates found to be associated with missingness at post- or follow-up were included in the model to account for data that were not missing completely at random.

We conducted a series of three-level, multilevel models (Heck & Thomas, 2015) in *Mplus* (Muthén & Muthén, 1998–2017) to examine the impact of the intervention on growth in outcomes from baseline to one-year follow-up. A separate model was fit for each BASC-2 outcome (i.e., four teacher-reported composites and four student-reported composites). Repeated measures were accounted for at level one while student- and school-level associations were modeled at level two and level three, respectively. Full-information robust maximum likelihood was utilized to account for multivariate non-normality in the data as well as missing data that were assumed to be missing at random (Enders & Bandalos, 2001; Muthén & Asparouhov, 2002). Covariates were the same in all final teacher- and student reported outcome models. At level one (repeated measures), we estimated two random effects as growth over time:

$$Y_{ijt} = \pi_{0ji} + \pi_{1ji}T_{ijt} + \varepsilon_{ijt} \quad (1)$$

Based on the linear time steps within T_{ijt} , we estimated a random intercept (π_{0ji}), which is the estimated, conditional mean value of the outcome as measured at baseline, and a random slope (π_{1ji}), the estimated per unit time change in Y . These values varied across individual i within school j .

At Level 2 (student-level), the Level-1 intercept and time slope are outcomes:

$$\pi_{pij} = \beta_{p0j} + \sum_{q=1}^{Q_p} \beta_{qpj}X_{qji} + r_{pij} \quad (2)$$

B_{qpj} is a matrix of coefficients corresponding to the effect of between-individual predictor q on Level-1 outcome p (intercept, slope) within school j . The key predictors included in X_{qji} were student sex (female = 1; male = 0), race/ethnicity (Black/African American = 1; Other = 0), and cohort; these variables were included as predictors of the random intercept and random slope of linear growth. In addition, within each set of reporter-specific outcomes (i.e., teacher-reported composites, student-reported composites), baseline BASC-2 composites were included as covariates at level two (e.g., when modeling a teacher-reported BASC-2 outcome, the other teacher-reported composites were included as covariates). All student-level predictors were group mean centered.

School-level predictors were included at Level 3, including the focal ITT parameter (i.e., EACP vs. school as usual; see Schuelke et al., 2013):

$$\beta_{pqj} = \gamma_{pq0} + \sum_{s=1}^{S_{pq}} \gamma_{pqs}Z_{sj} + u_{pqj} \quad (3)$$

Z_{sj} contains a series of school-level predictors that may account for variation across schools. In addition to school intervention status (i.e., EACP vs. school as usual), aggregated school means of each student-level predictor were included as predictors and grand mean centered. School site (i.e., state of implementation) was also included. After fitting the model with all predictors, we calculated an effect size (i.e., $d_{GMA-RAW}$; Feingold, 2009; Raudenbush & Liu, 2001) as the difference in growth trajectories for EACP versus school as usual from baseline to follow-up, divided by the standard deviation of the outcome at baseline, which is equivalent to a Cohen's d .

After estimating the ITT models, a final step in the model-building process explored modification of intervention effects by student sex and race, as well as a baseline by intervention interaction. To explore effect modification (i.e., cross-level interaction on growth) by sex and race, we estimated two models for each BASC-2 outcome (i.e., one to explore effect modification by sex and one for race). We expanded on the model where student sex and race had a fixed effect on growth over time (as denoted in Eq. 2) and estimated a random effect of student sex or race, such that the impact of student sex or race on growth was free to vary across schools. We explored the cross-level interactions of interest by specifying the ITT dummy variable at level three as a predictor of this random effect of race or sex

on growth.

To examine baseline by intervention interactions, we reparametrized the model as a latent growth model under the specifications of Muthén and Curran (1997; Fig. 5) in a series of multigroup models (i.e., EACP schools vs. school-as-usual schools). The models were extended to accommodate the additional nesting of students within schools (i.e., two-level latent growth models). In the school-as-usual group, we modeled linear growth in the outcome using an initial status and growth latent factor at the within-school and between-school levels. In the EACP group, in addition to the initial status and growth factors, an additional growth factor was estimated which represents the additional growth over time that is attributable to EACP. The covariances between the initial status factor and additional slope parameter in EACP schools at the within- and between-levels represents the baseline by intervention interaction. The significance of these terms was determined by the respective Wald tests ($p < .05$). All covariates were included as indicated above.

6. Results

6.1. Intervention effect on social, emotional, behavior and adaptive outcomes

6.1.1. Teacher-reported BASC-2 outcomes

Consistent with our primary outcomes of interest, we first examined the impacts on the teacher-reported BASC-2 scores, which included the potential for exploratory cross-over effects on Internalizing Problems. The results of the final three-level models for teacher-reported BASC-2 outcomes are presented in Table 3. The table denotes estimates of both the model intercept and slope parameters and the estimated ITT effect of EACP on the model intercept and slope. We also denote a Cohen's d effect size for each main effect. EACP was associated with differences in linear growth trajectories for one of the four teacher-reported outcomes. EACP was associated with a statistically significant decrease in Externalizing Problems over time ($\gamma_{EACP} = -1.36, p = .049, d_{GMA-RAW} = -0.21$). EACP was not significantly associated with Internalizing Problems, Adaptive Skills, or School Problems growth trajectories.

An additional set of analyses on the teacher outcomes explored effect modification; these findings suggested a statistically significant interaction between sex and intervention on School Problems ($\gamma_{EACP*Sex} = -1.57, p = .035$), such that girls in the EACP condition demonstrated decreases in School Problems over time. However, there was not a statistically significant interaction between intervention condition and race ($\gamma_{EACP*Race} = -0.64, p = .522$). Additional effect modification analyses did not suggest statistically significant cross-level interactions between intervention status and sex or intervention status and race on Externalizing Problems ($\gamma_{EACP*Sex} = -0.57, p = .510$; $\gamma_{EACP*Race} = 0.44, p = .605$), Internalizing Problems ($\gamma_{EACP*Sex} = -0.58, p = .509$; $\gamma_{EACP*Race} = 0.02, p = .976$), or Adaptive Skills ($\gamma_{EACP*Sex} = 1.41, p = .118$; $\gamma_{EACP*Race} = 0.59, p = .525$). Similarly, there were no significant baseline by intervention interactions on Externalizing Problems ($\gamma_{Baseline*ITT} = -0.19, p = .102$; $\beta_{Baseline*ITT} = -0.08, p = .097$), Adaptive Skills ($\gamma_{Baseline*ITT} = 0.03, p = .785$; $\beta_{Baseline*ITT} = 0.04, p = .346$), or School Problems ($\gamma_{Baseline*ITT} = -0.03, p = .823$; $\beta_{Baseline*ITT} = -0.02, p = .764$). However, there was a significant baseline by intervention interaction on Internalizing Problems ($\gamma_{Baseline*ITT} = -0.47, p = .036$; $\beta_{Baseline*ITT} = -0.15, p < .001$), such that students with greater levels of Internalizing Problems at baseline and in schools randomized to EACP experienced attenuated growth in Internalizing Problems over time and students in schools with greater levels of Internalizing Problems at baseline and in schools randomized to EACP experienced greater attenuation in growth in Internalizing Problems over time. In this model, the main effect of the intervention dropped to non-significance ($\gamma_{EACP} = -1.12, p = .075$).

Table 3

Three-level models examining the intent to treat impact of EACP on teacher- and student-reported BASC-2 composites.

Outcome	Parameter	Coefficient	<i>p</i>	ITT	<i>p</i>	<i>d</i>
<i>Teacher-reported BASC – 2 composites (T-scores)</i>						
Externalizing Problems	Intercept (γ_{000})	62.85	< 0.001	1.18	0.378	
	Slope (γ_{100})	1.00	0.130	−1.36	0.049	−0.21
Internalizing Problems	Intercept (γ_{000})	49.76	< 0.001	2.82	0.012	
	Slope (γ_{100})	2.12	0.006	−1.73	0.073	−0.33
Adaptive Skills	Intercept (γ_{000})	41.80	< 0.001	−1.05	0.104	
	Slope (γ_{100})	0.65	0.237	0.88	0.095	0.21
School Problems	Intercept (γ_{000})	58.01	< 0.001	−0.38	0.585	
	Slope (γ_{100})	0.13	0.819	−0.47	0.396	−0.10
<i>Student-reported BASC – 2 composites (T-scores)</i>						
School Problems	Intercept (γ_{000})	51.99	< 0.001	0.52	0.431	
	Slope (γ_{100})	0.33	0.300	−0.04	0.904	−0.01
Internalizing Problems	Intercept (γ_{000})	52.21	< 0.001	−1.34	0.029	
	Slope (γ_{100})	−0.47	0.376	0.70	0.045	0.13
Inattention/ Hyperactivity	Intercept (γ_{000})	52.61	< 0.001	−0.02	0.981	
	Slope (γ_{100})	0.71	0.117	−0.04	0.932	−0.01
Personal Adjustment	Intercept (γ_{000})	51.19	< 0.001	−0.71	0.295	
	Slope (γ_{100})	−0.70	0.109	−0.10	0.824	−0.02

Note. Behavior Assessment System for Children (BASC). The coefficient column denotes the model estimated intercept or slope value. Bold indicates significant at $p < .05$.

6.1.2. Student-reported BASC-2 outcomes

Our second set of outcomes related to the exploratory analyses regarding the student-reported BASC-2 externalizing and potential cross-over Internalizing Problems. Results (reported in Table 3) indicated that EACP was associated with a small, but statistically significant increase in Internalizing Problems over time ($\gamma_{EACP} = 0.70, p = .045, d_{GMA-RAW} = 0.13$). There were no other significant ITT effects of EACP on growth in student-reported BASC-2 outcomes. Additional analyses examining effect modification resulted in one statistically significant cross-level interaction between intervention status and sex ($\gamma_{EACP*Sex} = 1.83, p = .048$), which indicated that girls in EACP demonstrated more favorable self-reported Personal Adjustment outcomes. There was not a significant cross-level interaction of intervention status and race on student-report Personal Adjustment ($\gamma_{EACP*Race} = -1.15, p = .227$). Additional analyses examining effect modification did not suggest statistically significant cross-level interactions between intervention status and sex or intervention status and race on School Problems ($\gamma_{EACP*Sex} = 1.70, p = .051$; $\gamma_{EACP*Race} = 0.31, p = .756$), Internalizing Problems ($\gamma_{EACP*Sex} = -0.78, p = .295$; $\gamma_{EACP*Race} = -0.10, p = .944$), or Inattention/Hyperactivity ($\gamma_{EACP*Sex} = 0.12, p = .887$; $\gamma_{EACP*Race} = -1.30, p = .245$). Similarly, there were no significant baseline by intervention interactions on School Problems ($\gamma_{Baseline*ITT} = 0.06, p = .738$; $\beta_{Baseline*ITT} = 0.01, p = .939$), Internalizing Problems ($\gamma_{Baseline*ITT} = -0.05, p = .830$; $\beta_{Baseline*ITT} = 0.02, p = .693$), Inattention/Hyperactivity ($\gamma_{Baseline*ITT} = -0.11, p = .710$; $\beta_{Baseline*ITT} = 0.00, p = .949$), or Personal Adjustment ($\gamma_{Baseline*ITT} = -0.34, p = .692$; $\beta_{Baseline*ITT} = -0.08, p = .449$).

6.2. Sensitivity analyses: inverse probability of intervention weighting

We also conducted a series of sensitivity analyses related to baseline imbalance across EACP and school-as-usual schools (see Supplementary Tables). Specifically, the sensitivity analyses added inverse probability of intervention weights (IPTW) to the primary intent-to-treat models. Model-specific, school level weights were created by estimating the probability of intervention assignment conditional on baseline values of the focal BASC-2 outcome. The inverse was utilized as a school-level weight in the three-level ITT models. Covariates were included in a penultimate model, as described above but retained only if significantly related to growth. Taken together, the results of the sensitivity analyses were largely consistent with the results presented above, particularly with regard to the ITT point estimates. These in turn give us more confidence in the overall pattern of findings, despite the baseline imbalance on the two teacher-reported BASC-2 scores.

7. Discussion

This study aimed to contribute to the knowledge base on Coping Power, as there has been limited examination of the impacts of this and other prevention programs during the middle school years (Alperin et al., 2023; Evans et al., 2007). These results suggest that the structured social-cognitive indicated prevention program, which includes youth, caregiver, and teacher elements, can be adapted for use with early adolescents, and have a significant main effect on teacher ratings of externalizing problems through a 1-year follow up. Notably, 8th grade teachers, who completed the follow-up assessment were unaware of the intervention condition. There was also an interaction between sex and intervention on School Problems over time, such that girls in EACP demonstrated decreases in School Problems over time.

The intervention effect on teacher-reported adolescent externalizing behavior is notable because of the relative dearth of indicated school-based, evidence-based interventions for youth with difficulties regulating aggression during the middle school years, especially in relation to the growing intervention literature for similar children in the preadolescent years (Alperin et al., 2023; Daly et al., 2018; Evans et al., 2007). Although a number of rigorous studies have demonstrated that classroom-level universal prevention in the middle school years can have significant positive effects on youth substance use, antisocial behavior, aggression and bullying (e.g., Botvin et al., 2006; Espelage et al., 2015; Marshall et al., 2024) the effects of multicomponent interventions targeting early adolescents with aggressive behavior problems has been limited and sometimes discouraging. For example, the comprehensive multicomponent Fast Track program that provided school-based intervention to children with behavior problems and their families in a sustained way, from first grade through tenth grade, had its significant positive effects evident during elementary school (Conduct Problems Prevention Research Group, 2002, 2004) fade dramatically during the middle school years (Conduct Problems Prevention Research Group, 2010), before re-emerging again in later adolescence and young adulthood (Conduct Problems Prevention Research Group, 2015). The difficulty in obtaining stable positive effects in targeted interventions with at-risk adolescents in middle school may be due to the dramatic changes that occur during adolescence in youths' hormonal and self-regulatory processes, and in their peer relations which can be increasingly influenced by deviant peer associations (Utrzan et al., 2018). These changes in self-regulatory and peer processes can negatively affect youth response to small group interventions (e.g., Dishion & Andrews, 1995) and to their behavioral development. In this context, then, it is encouraging that EACP can help reduce the escalation of externalizing behaviors among at-risk students during this developmental period; moreover, these effects are apparent for boys and girls, and for Black and White students. This developmentally adapted version of Coping Power likely reduces the students' externalizing problems by addressing the developmentally salient emotional regulation, decision-making, and peer relationship skills and the caretakers' monitoring and consistent discipline skills.

The student-self-reported outcomes were exploratory and secondary to the teacher-reported outcomes because this was among the first study of Coping Power to explore self-reported outcomes. In fact, youth self-reports of externalizing behavior have not been found to be as related to occurrence of externalizing disorders as are teacher and parent/caregiver reports (Salbach-Andrae et al., 2009), and thus child behavioral reports may not be as predictive to their later serious behavioral outcomes as are teacher reports. Moreover, teacher reports may be more reliable sources of information on students' behavior in the school setting, whereas students may lack

insight on the impact of their behavior in relation to others, especially in the school context. Furthermore, teachers are natural raters and their perspectives of students' behavior play an important role in the education context (for further discussion of discrepancies in reporters, see [De Los Reyes et al., 2023](#)). The results suggested that EACP was associated with a small, statistically significant increase in Internalizing Problems over time, although not leading to overall mean scores in the clinical range. Some children with externalizing problems have low levels of subjective anxiety, are less concerned about consequences of their behavior and thus have poorer outcomes ([Fanti et al., 2016](#); [Frick & Wall Myers, 2018](#)). Modest levels of anxiety and depression have been associated with better intervention response among children with disruptive behaviors ([Jarrett et al., 2014](#); [Kazdin & Whitley, 2006](#); [Ollendick et al., 2008](#)), and, in the present study, modest increases in Internalizing Problems may indicate children's increased awareness of consequences of externalizing behaviors. More research to replicate this finding, including measures on potential mechanisms (e.g., increased insight and awareness), would benefit the field.

In addition to EACP effects on the primary outcome of this intervention, namely externalizing behavior problems, EACP also reduced the escalation of teacher-reported internalizing problems among adolescents with high baseline levels of internalizing problems, and among those attending schools with at-risk peers who also had high rates of internalizing behaviors. Comorbidity of externalizing and internalizing symptoms can be evident among a subgroup of aggressive youth. Students who have an angry-irritable subtype of oppositional behaviors can have comorbid depression symptoms both concurrently and longitudinally, likely due to underlying emotion regulation difficulties, and be at risk for future suicidal behaviors ([Evans et al., 2017](#)). Those with both conduct problems and depressive symptomatology are also more likely to engage in substance use as adolescents than those children with conduct problems alone ([Miller-Johnson et al., 1998](#)). While EACP may be considered a disorder-specific intervention given its focus on children with aggressive behavior issues, it may contain certain transdiagnostic elements that impact other presenting symptoms along with externalizing behaviors ([Clifford et al., 2020](#)). In fact, similar to outcomes from other interventions for preadolescents (e.g., [Chase & Eyberg, 2008](#); [Kazdin, 2010](#)), research has shown that Coping Power delivered in the elementary school years can produce reductions in both externalizing and internalizing behaviors ([Lochman et al., 2015](#)), and has been particularly effective in reducing externalizing behaviors for students who have higher baseline depression ([Jarrett et al., 2014](#)). The current findings extend these transdiagnostic intervention effects to Coping Power as adapted and implemented during the middle school years.

EACP may have these effects either because (a) some internalizing problems follow youth's development of aggressive behaviors and therefore changes in aggression may influence these outcomes, or (b) there are some common active mechanisms within the program that may affect multiple areas ([Clifford et al., 2020](#); [Lochman et al., 2019](#)). EACP appears to target a number of common risk factors that predict both externalizing and internalizing problems, including psychosocial stressors in varied environmental contexts (school, neighborhood, peer group), difficult temperament (e.g., irritability, restlessness, lack of persistence, low adaptability), social-cognitive distortions, and family factors such as harsh discipline practices, low levels of parental warmth and acceptance, multiple changes in family composition, and parental psychopathology. EACP addresses these risk factors with transdiagnostic intervention activities, including personal goal setting, emotional awareness and regulation, perspective-taking, social problem solving, and rehearsal opportunities.

In addition, there was a significant intervention impact for girls on their self-reported Personal Adjustment. Adolescents' self-ratings of the quality of their personal relationships, and their levels of self-esteem and self-reliance comprised the BASC composite for personal adjustment. Female adolescents assigned to EACP appeared to have reported growth in their personal adjustment likely related to the self-regulatory, social problem solving, and decision-making EACP activities. This finding appears consistent with Coping Power effects in the elementary school years, as children receiving Coping Power, in comparison to control conditions, have been found to have higher levels of perceived social competence, and an improved internal locus of control, and expectations that aggressive solutions to interpersonal problems will not lead to positive social outcomes ([Lochman & Wells, 2002a, 2002b](#)). Intervention-induced improvements in youth personal adjustment may assist the youth to sustain their observed behavior changes and affect their long-term behavioral and emotional outcomes.

It is noteworthy that given the older age group enrolled in the EACP, we were able to examine student-reported outcomes, whereas other studies of Coping Power to date have predominately focused on teacher and caregiver report. Although teachers observed and reported significant intervention effects for EACP on externalizing symptoms, in general, students' reports of their externalizing behavior were not affected by their assignment to EACP or to the control condition. Similarly, we observed an effect of EACP on teacher-reported Internalizing Problems, in the context of higher baseline levels of these challenges, but student reports showed that those in the EACP intervention had increased self-reported Internalizing Problems. Such discrepancies in ratings are not uncommon in the research ([De Los Reyes et al., 2023](#)). For example, [Chen et al. \(2009\)](#) examined externalizing behavior scores from teachers, adolescent students and peers on Frick's Clustering Child Behavior Checklist and when the three sources were compared, the lowest correlations were between teacher and student, and the highest were between teachers and peers (so both were behavior observers). Taken together, these studies raise questions about the validity of student report symptoms over teacher ratings, leading us to put more weight on the teacher ratings than youth self-report, as the teachers may have a better sense of the students' behavior than the students' self-ratings, particularly the externalizing symptoms. Additional research is needed to better understand the sources of discrepancies by respondents ([De Los Reyes et al., 2023](#)).

7.1. Limitations and future directions

We report findings from a large trial with 709 students across 40 schools, with randomization conducted at the school-level. Despite the randomized design, baseline equivalence tests suggested some imbalance at baseline; specifically, there was baseline inequivalence on two of the teacher-reported BASC subscales, but none of the other ten scales. Baseline inequivalence is a common source of bias in

trials of preventive interventions (Steege et al., 2021) and may be particularly problematic in group randomized trials; however, as noted in the results section, many of the intercept differences in those models fell to statistical non-significance. However, in the multilevel ITT models, the baseline differences between EACP and school as usual on teacher-reported Internalizing Problems persisted and there was a significant baseline difference denoted for student-reported Internalizing Problems. Sensitivity analyses including IPTW suggested a similar pattern of results, yet the main effect of EACP on teacher-reported Externalizing Problems fell below the $p < .05$ cutoff, while the point estimate was similar to the primary ITT effect ($\gamma_{EACP} = -1.35$, $p = .076$, $d_{GMA-RAW} = -0.21$). Additionally, the EACP impact on teacher-reported Internalizing Problems in the IPTW models was statistically significant ($\gamma_{EACP} = -1.92$, $p = .032$, $d_{GMA-RAW} = -0.37$). However, this impact is likely best characterized in the context of the baseline by intervention interaction.

We did not adjust for multiple comparisons in the final outcome analyses. We noted that emerging theoretical and empirical evidence by Clifford et al. (2020) suggests that Coping Power may have transdiagnostic impacts on internalizing problems. As such, we organized our primary hypotheses to focus on teacher-reported externalizing problems, and our secondary outcomes related to teacher- and student-reported symptoms, with this latter set of outcomes being largely exploratory, as were the internalizing outcomes and the effect modification analyses. Therefore, we did not apply more conservative corrections like Bonferroni for the multiple tests (see Schochet, 2009 for a framework for applying corrections for multiple comparisons in impact evaluations).

Differential attrition was observed by intervention condition and other covariates. Differential attrition is a common concern in RCTs of preventive interventions that can lead to bias (Steege et al., 2021). We utilized FIML to account for missing data and included all covariates associated with attrition (including intervention condition), which should produce unbiased estimates of the intervention effects under the MAR assumption (Enders & Bandalos, 2001; Steeger et al., 2021). Given the robust array of covariates included at the student and school level across all models, we believe that the MAR assumption is tenable.

The first gate of the screening process for this study involved teacher ratings of students' aggressive behaviors. Although teacher ratings of children's behavior problems have been found to be valid in predicting students' long-term outcomes, and to be superior to parent ratings (Hill et al., 2004; Kassing et al., 2019; Whipp et al., 2019), the potential for teacher bias in behavior ratings is a limitation. Teacher race has been found to affect their ratings of children's behavior from the preschool years (Wymer et al., 2022) through the later school years (Redding, 2019), with White teachers rating Black children as having higher rates of externalizing behavior problems than do Black teachers. There is also evidence that teachers' misperceptions of Black children's behavior can be influenced by implicit biases, and teachers' racially-biased perceptions of children's disruptive behavior can lead to Black children receiving disproportionate discipline (Girvan et al., 2017). We did not collect demographic data on teachers, as they were not enrolled as participants in the project; rather, they were considered raters of student outcomes and, in some cases, helped to deliver EACP program content. As a result, we lack data to examine teacher biases in screening. Moreover, there is little research examining whether and how such data should be used to adjust screening data for eligibility to participate in similar interventions. More research is needed to inform the approach to be taken adjusting for teacher-student racial mismatch and potential teacher racial bias in future studies using a similar screening approach.

Strength of the study include the large sample and random assignment across schools to deal with the potential of contamination; while it is possible that 6th grade teachers conducting the screening or the 8th grade teachers conducting the follow-up ratings may have become aware of the schools intervention condition over time, this seems unlikely given the relatively small number of students participating in the intervention and the fact they had no other interaction with the program components. It is possible that the 7th grade teachers who conducted the baseline and end of Year 1 BASC ratings were aware of the schools' intervention status over time, which could have influenced their ratings; some of these teachers were also "cheer teachers" in the intervention schools.

The findings presented here focused on the ITT analyses, and results are based on project hired implementers, at one-year follow-up. Future analyses will explore additional subgroup effects, as well as variation based on additional baseline characteristics (e.g., academic challenges) and possible mediators (e.g., social-cognitive changes). We also intend to explore other possible mechanisms of the change process, like the specific social cognitive factors consistent with the theory of change process as originally theorized and outlined by Lochman and Wells (2003). Examination of other outcomes like academic performance and discipline problems, including suspensions and attendance, will be conducted once data are obtained from the participating school divisions. The overall fidelity of implementation was high for students (Pas et al., 2020), although there was more variability in caregiver attendance, which could prove to be an important effect modifier to explore in future studies, as it has been shown to be in prior Coping Power studies (Lindsey et al., 2019); this is a factor to be explored in relation to student outcomes in future studies.

8. Conclusion

Our findings suggest that EACP is a promising model for addressing early adolescents' mental and behavioral health needs in schools. Of particular interest are the potential crossover effects on internalizing symptoms, which highlight the transdiagnostic utility of EACP. Given the increasing need for programs to help early adolescents bridge the middle school transition, especially in light of elevated challenges during the COVID-19 pandemic recovery and the escalating youth mental health crisis (OSG, 2021; Weist et al., 2024), EACP holds promise for helping school psychologists meet some of the growing needs for evidence-based indicated interventions in middle schools and overcoming barriers to services.

CRedit authorship contribution statement

Catherine P. Bradshaw: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Heather McDaniel:** Writing – review & editing,

Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Elise T. Pas:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Katrina J. Debnam:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Jessika H. Bottiani:** Writing – original draft, Project administration. **Nicole Powell:** Supervision, Investigation, Conceptualization. **Nicholas S. Ialongo:** Writing – review & editing, Investigation, Conceptualization. **Antonio Morgan-Lopez:** Methodology, Formal analysis. **John E. Lochman:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Funding acquisition, Conceptualization.

Acknowledgement

We would like to acknowledge the clinicians who supported implementation of the trial, including Dr. Lauren Kaiser and Christina Knepper at the Maryland site, as well as Shane Jones at the Alabama site. We also thank Dr. Amanda Nguyen, Qing Zheng, and Lixin Qu for data management support. This research was supported in part with funding from the Institute of Education Sciences, U.S. Department of Education, through grant R305A140070 (PI: C. Bradshaw) to the University of Virginia, and the National Institute of Mental Health (P30 MH086043; PI N. Ialongo) to the Johns Hopkins Bloomberg School of Public Health. This content is solely the responsibility of the authors and does not represent the official views of the Institute of Education Sciences or Department of Education, or the National Institute of Mental Health.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jsp.2025.101437>.

References

- Alperin, A., Reddy, L. A., Glover, T. A., Bronstein, B., Wiggs, N. B., & Dudek, C. M. (2023). School-based interventions for middle school students with disruptive behaviors: A systematic review of components and methodology. *School Psychology Review*, 52(2), 180–205. <https://doi.org/10.1080/2372966X.2021.1883996>.
- Botvin, G. J., Griffin, K. W., & Nichols, T. R. (2006). Preventing youth violence and delinquency through a universal school-based prevention approach. *Prevention Science*, 7(4), 403–408. <https://doi.org/10.1007/s11121-006-0057-y>.
- Bradshaw, C. P., Lochman, J. E., Powell, N., & Ialongo, N. (2017). Preventing bullying in middle schoolers by using the Coping Power Program: A targeted group intervention. In C. P. Bradshaw (Ed.), *Handbook on bullying prevention: A life course perspective* (pp. 215–220). NASW Press.
- Chase, R. M., & Eyberg, S. M. (2008). Clinical presentation and treatment outcome for children with comorbid externalizing and internalizing symptoms. *Journal of Anxiety Disorders*, 22(2), 273–282. <https://doi.org/10.1016/j.janxdis.2007.03.006>.
- Chen, G. H., Zhang, W. X., & Wang, S. Q. (2009). Agreement among different informants over ratings of adolescent externalizing behaviors. *Acta Psychologica Sinica*, 41(2), 144–155. <https://doi.org/10.3724/SP.J.1041.2009.00144>.
- Clifford, M. E., Nguyen, A. J., & Bradshaw, C. P. (2020). Both/and: Tier 2 interventions with transdiagnostic utility in addressing emotional and behavioral disorders in youth. *Journal of Applied School Psychology*, 36(2), 173–197. <https://doi.org/10.1080/15377903.2020.1714859>.
- Conduct Problems Prevention Research Group. (2002). Evaluation of the first three years of the Fast Track prevention trial with children at high risk of adolescent conduct problems. *Journal of Abnormal Child Psychology*, 30(1), 19–35. <https://doi.org/10.1023/A:1014274914287>.
- Conduct Problems Prevention Research Group. (2004). The effects of the Fast Track program on serious problem outcomes at the end of elementary school. *Journal of Clinical Child and Adolescent Psychology*, 33(4), 650–661. https://doi.org/10.1207/s15374424jccp3304_1.
- Conduct Problems Prevention Research Group. (2010). The difficulty of maintaining positive intervention effects: A look at disruptive behaviors, deviant peer relations, and social skills during the middle school years. *The Journal of Early Adolescence*, 30(4), 593–624. <https://doi.org/10.1177/0272431609340513>.
- Conduct Problems Prevention Research Group. (2015). Impact of early intervention on psychopathology, crime, and well-being at age 25. *American Journal of Psychiatry*, 172(1), 59–70. <https://doi.org/10.1176/appi.ajp.2014.13060786>.
- Daly, B. P., DeMatteo, D., Hildenbrand, A., Baker, C. N., & Fisher, J. H. (2018). Psychosocial treatment and prevention in the adolescent years for ODD and CD. In J. E. Lochman, & W. Matthys (Eds.), *The Wiley handbook of disruptive and impulse-control disorders* (pp. 467–483). Hoboken, NJ: John Wiley & Sons. <https://doi.org/10.1002/9781119092254.ch28>.
- De Los Reyes, A., Epkins, C. C., Asmundson, G. J. G., Augenstein, T. M., Becker, K. D., Becker, S. P., ... Youngstrom, E. A. (2023). Editorial statement about JCCAP's 2023 special issue on informant discrepancies in youth mental health assessments: Observations, guidelines, and future directions grounded in 60 years of research. *Journal of Clinical Child and Adolescent Psychology*, 52(1), 147–158. <https://doi.org/10.1080/15374416.2022.2158842>.
- Dishion, T. J., & Andrews, D. W. (1995). Preventing escalation in problem behaviors with high-risk adolescents: Immediate and 1-year outcomes. *Journal of Consulting and Clinical Psychology*, 63(4), 538–548. <https://doi.org/10.1037/0022-006X.63.4.538>.
- Dodge, K. A., Lochman, J. E., Harnish, J. D., Bates, J. E., & Pettit, G. S. (1997). Reactive and proactive aggression in school children and psychiatrically-impaired chronically-assaultive youth. *Journal of Abnormal Psychology*, 106(1), 37–51. <https://doi.org/10.1037/0021-843X.106.1.37>.
- Eccles, J. S., Lord, S., & Buchanan, C. M. (2018). School transitions in early adolescence: What are we doing to our young people?. In *Transitions through Adolescence* (pp. 251–284). Psychology Press. <https://doi.org/10.4324/9781315789286>.
- Enders, C. K., & Bandalos, D. L. (2001). The relative performance of full information maximum likelihood estimation for missing data in structural equation models. *Structural Equation Modeling*, 8(3), 430–457. https://doi.org/10.1207/S15328007SEM0803_5.
- Espelage, D. L., Polanin, J. R., & Rose, C. A. (2015). Social-emotional learning program to reduce bullying, fighting, and victimization among middle school students with disabilities. *Remedial and Special Education*, 36(5), 1–13. <https://doi.org/10.1177/0741932514564564>.
- Evans, S., Owens, J., Bradshaw, C. P., & Weist, M. D. (Eds.). (2023). *Handbook of school mental health: innovations in science and practice* (3rd ed.). New York: Springer.
- Evans, S. C., Burke, J. D., Roberts, M. C., Fite, P. J., Lochman, J. E., de la Pena, F. R., & Reed, G. M. (2017). Irritability in child and adolescent psychopathology: An integrative review for ICD-11. *Clinical Psychology Review*, 53, 29–45. <https://doi.org/10.1016/j.cpr.2017.01.004>.
- Evans, S. W., Serpell, Z. N., Schultz, B. K., & Pastor, D. A. (2007). Cumulative benefits of secondary school-based treatment of students with attention deficit hyperactivity disorder. *School Psychology Review*, 36(2), 256–273. <https://doi.org/10.1080/02796015.2007.12087943>.
- Fanti, K. A., Panayiotou, G., Lazarou, C., Michael, R., & Georgiou, G. (2016). The better of two evils? Evidence indicating that children exhibiting continuous problems high and low on callous-unemotional traits score in opposite directions on physiological and behavioral measures of fear. *Development and Psychopathology*, 28(1), 185–198. <https://doi.org/10.1017/S0954579415000371>.

- Farmer, T. W., Irvin, M. J., Motoca, L. M., Leung, M. C., Hutchins, B. C., Brooks, D. S., & Hall, C. M. (2015). Externalizing and internalizing behavior problems, peer affiliations, and bullying involvement across the transition to middle school. *Journal of Emotional and Behavioral Disorders*, 23(1), 3–16. <https://doi.org/10.1177/1063426613491286>.
- Farrell, A. D., Goncey, E. A., Sullivan, T. N., & Thompson, E. L. (2018). Victimization, aggression, and other problem behaviors: Trajectories of change within and across middle school grades. *Journal of Research on Adolescence*, 28(2), 438–455. <https://doi.org/10.1111/jora.12346>.
- Feingold, A. (2009). Effect sizes for growth-modeling analysis for controlled clinical trials in the same metric as for classical analysis. *Psychological Methods*, 14(1), 43. <https://doi.org/10.1037/a0014699>.
- Frick, P. J., & Wall Myers, T. D. (2018). Conduct Disorder and callous-unemotional traits. In J. E. Lochman, & W. Matthys (Eds.), *The Wiley handbook of disruptive and impulse-control disorders* (pp. 37–54). Hoboken NJ: John Wiley and Sons. <https://doi.org/10.1002/9781119092254.ch3>.
- Girvan, E. J., Gion, C., McIntosh, K., & Smolkowski, K. (2017). The relative contribution of subjective office referrals to racial disproportionality in school discipline. *School Psychology Quarterly*, 32(3), 392–404. <https://doi.org/10.1037/spq0000178>.
- Heck, R. H., & Thomas, S. L. (2015). *An introduction to multilevel modeling techniques: MLM and SEM approaches using Mplus* (3rd ed.). Routledge/Taylor & Francis Group <https://doi.org/10.4324/9781315746494>.
- Herman, K. C., Reinke, W. M., Bradshaw, C. P., Lochman, J. E., Boxmeyer, C. L., Powell, N. P., ... Jalongo, N. S. (2012). Integrating the family check-up and the parent coping power program. *Advances in School Mental Health Promotion*, 5(3), 208–219. <https://doi.org/10.1080/1754730X.2012.707437>.
- Herman, K. C., Reinke, W. M., Dong, N., & Bradshaw, C. P. (2022). Can effective classroom behavior management increase student achievement in middle school? Findings from a group randomized trial. *Journal of Educational Psychology*, 114(1), 144–160. <https://doi.org/10.1037/edu0000641>.
- Hill, L. G., Lochman, J. E., Coie, J. D., Greenberg, M. T., & Conduct Problems Prevention Research Group. (2004). Effectiveness of early screening for externalizing problems: Issues of screening accuracy and utility. *Journal of Consulting and Clinical Psychology*, 72(5), 809–820. <https://doi.org/10.1037/0022-006X.72.5.809>.
- Jarrett, M. A., Siddiqui, S., Lochman, J. E., & Qu, L. (2014). Internalizing problems as a predictor of change in externalizing problems in at-risk youth. *Journal of Clinical Child and Adolescent Psychology*, 43(1), 27–35. <https://doi.org/10.1080/15374416.2013.764823>.
- Kassing, F., Godwin, J., Lochman, J. E., Coie, J. D., & Conduct Problems Prevention Research Group. (2019). Using early childhood behavior problems to predict adult convictions. *Journal of Abnormal Child Psychology*, 47, 765–778. <https://doi.org/10.1007/s10802-018-0478-7>.
- Kazdin, A., & Whitley, M. (2006). Comorbidity, case complexity, and effects of evidence-based treatment for children referred for disruptive behavior. *Journal of Consulting and Clinical Psychology*, 74(3), 455–467. <https://doi.org/10.1037/0022-006X.74.3.455>.
- Kazdin, A. E. (2010). Problem-solving skills training and parent management training for oppositional defiant disorder and conduct disorder. In J. R. Weisz, & A. E. Kazdin (Eds.), *Evidence-based psychotherapies for children and adolescents* (2nd ed., pp. 211–226). New York: Guilford Press.
- Lindsey, M., Romanelli, M., Ellis, M., Barker, E., Boxmeyer, C., & Lochman, J. (2019). The influence of treatment engagement on positive outcomes in the context of a school-based intervention for students with externalizing behavior problems. *Journal of Abnormal Child Psychology*, 47(2), 1437–1454. <https://doi.org/10.1007/s10802-019-00525-6>.
- Lochman, J. E., Boxmeyer, C. L., Kassing, F. L., Powell, N. P., & Stromeyer, S. L. (2019). Cognitive-behavioral intervention for youth at-risk for conduct problems: Future directions. *Journal of Clinical Child and Adolescent Psychology*, 48(5), 799–810. <https://doi.org/10.1080/15374416.2019.1567349>.
- Lochman, J. E., Boxmeyer, C. L., Powell, N. P., Qu, L., Wells, K., & Windle, M. (2012). Coping power dissemination study: Intervention and special education effects on academic outcomes. *Behavioral Disorders*, 37(3), 192–205. <https://doi.org/10.1177/019874291203700306>.
- Lochman, J. E., & Conduct Problems Prevention Research Group. (1995). Screening of child behavior problems for prevention programs at school entry. *Journal of Consulting and Clinical Psychology*, 63(4), 549–559. <https://doi.org/10.1037/0022-006X.63.4.549>.
- Lochman, J. E., Dishion, T. J., Powell, N. P., Boxmeyer, C. L., Qu, L., & Sallee, M. (2015). Evidence-based preventive intervention for preadolescent aggressive children: One-year outcomes following randomization to group versus individual delivery. *Journal of Consulting and Clinical Psychology*, 83(4), 728–735. <https://doi.org/10.1037/ccp0000030>.
- Lochman, J. E., Powell, N. P., & Jones, S. (2021). The Coping Power Program for children with aggressive behavior problems. In P. C. Kendall (Ed.), *Flexibility within fidelity* (pp. 211–230). New York, NY: Oxford University Press. <https://doi.org/10.1093/med-psych/9780197552155.001.0001>.
- Lochman, J. E., & Wells, K. C. (2002a). Contextual social-cognitive mediators and child outcome: A test of the theoretical model in the Coping Power program. *Development and Psychopathology*, 14(4), 945–967. <https://doi.org/10.1017/S0954579402004157>.
- Lochman, J. E., & Wells, K. C. (2002b). The Coping Power program at the middle-school transition: universal and indicated prevention effects. *Psychology of Addictive Behaviors*, 16(4S), S40–S54. <https://doi.org/10.1037/0893-164X.16.4S.S40>.
- Lochman, J. E., & Wells, K. C. (2003). Effectiveness of the coping power program and of classroom intervention with aggressive children: Outcomes at a 1-year follow-up. *Behavior Therapy*, 34(4), 493–515. [https://doi.org/10.1016/S0005-7894\(03\)80032-1](https://doi.org/10.1016/S0005-7894(03)80032-1).
- Lochman, J. E., & Wells, K. C. (2004). The Coping Power Program for preadolescent aggressive boys and their parents: Outcome effects at the 1-year follow-up. *Journal of Consulting and Clinical Psychology*, 72(4), 571–578. <https://doi.org/10.1037/0022-006X.72.4.571>.
- Lochman, J. E., Wells, K. C., & Lenhart, L. A. (2008). *Coping power: Child group program workbook*. Oxford University Press. <https://doi.org/10.1093/med:psych/9780195327861.001.0001>.
- Lochman, J. E., Wells, K. C., Qu, L., & Chen, L. (2013). Three year follow-up of Coping Power intervention effects: Evidence of neighborhood moderation? *Prevention Science*, 14(4), 364–376. <https://doi.org/10.1007/s11121-012-0295-0>.
- Macgowan, M. J. (1997). A measure of engagement for social group work: The Groupwork Engagement Measure (GEM). *Journal of Social Service Research*, 23(2), 17–37. https://doi.org/10.1300/J079v23n02_02.
- Marshall, T., Hoey, T., Rao, N., Taylor, J., George, P., Ghose, S. S., Cosgrove, J., & Patel, N. A. (2024). Reducing disruptive and distracting behaviors in the classroom: Assessing the evidence base. *Psychiatric Services*, 75(9). <https://doi.org/10.1176/appi.ps.20230543>.
- McDaniel, S. C., Watkins, L., Chow, J. C., Fedewa, M., & Nemer, S. (2023). Systematic literature review and meta-analysis of Coping Power: Effects and implications for implementation. *Journal of Applied School Psychology*, 39(4), 350–374. <https://doi.org/10.1080/15377903.2023.2196946>.
- Miller-Johnson, S., Lochman, J. E., Coie, J. D., Terry, R., & Hyman, C. (1998). Comorbidity of conduct and depressive problems at sixth grade: Substance use outcomes across adolescence. *Journal of Abnormal Child Psychology*, 26(3), 221–232. <https://doi.org/10.1023/A:1022676302865>.
- Monachino, C., Splett, J. W., Shen, Z., Cornett, S., Halliday, C. A., & Weist, M. D. (2021). Patterns and pathways of peer victimization across the transition to middle school. *School Psychology Review*, 50(2–3), 420–440. <https://doi.org/10.1080/2372966X.2021.1904792>.
- Murray, D. M. (1998). *Design and analysis of group-randomized trials*. Oxford Press.
- Muthén, B., & Asparouhov, T. (2002). Using Mplus Monte Carlo simulations in practice: A note on non-normal missing data in latent variable models. *Mplus Web Notes*, 2(2).
- Muthén, B. O., & Curran, P. J. (1997). General longitudinal modeling of individual differences in experimental designs: A latent variable framework for analysis and power estimation. *Psychological Methods*, 2(4), 371. <https://doi.org/10.1037/1082-989X.2.4.371>.
- Muthén, L. K., & Muthén, B. O. (1998–2017). *Mplus user's guide* (8th ed.). Los Angeles, CA: Muthén & Muthén.
- National Academies of Sciences, Engineering, and Medicine. (2019). *Fostering healthy mental, emotional, and behavioral development in children and youth: A national agenda*. Washington, DC: The National Academies Press. <https://doi.org/10.17226/25201>.
- National Center for Education Statistics (NCES). (2022). Discipline problems reported by public schools. In *Condition of education*. U.S. Department of Education, Institute of Education Sciences. Retrieved [1/15/24], from <https://nces.ed.gov/programs/coe/indicator/a07>.
- Office of the Surgeon General (OSG). (2021). *Protecting youth mental health: The U.S. Surgeon General's Advisory*. US Department of Health and Human Services. Retrieved from <https://www.hhs.gov/sites/default/files/surgeon-general-youth-mental-health-advisory.pdf>.
- Ollendick, T. H., Jarrett, M. A., Grills-Tauchel, A. E., Hovey, L. D., & Wolff, J. C. (2008). Comorbidity as a predictor and moderator of treatment outcome in youth with anxiety, affective, attention deficit/hyperactivity disorder, and oppositional/conduct disorders. *Clinical Psychology Review*, 28(8), 1447–1471. <https://doi.org/10.1016/j.cpr.2008.09.003>.

- Pas, E. T., Kaiser, L., Rabinowitz, J. A., Lochman, J. E., & Bradshaw, C. P. (2020). Patterns of student attendance for group counseling and the influence of student, group, and clinician factors. *Journal of Applied School Psychology*, 36(2), 198–226. <https://doi.org/10.1080/15377903.2020.1714860>.
- Pigott, T. D., & Polanin, J. R. (2015). The use of meta-analytic statistical significance testing. *Research Synthesis Methods*, 6(1), 63–73. <https://doi.org/10.1002/jrsm.1124>.
- Prado, G., Pantin, H., Schwartz, S. J., Lupei, N. S., & Szapocznik, J. (2006). Predictors of engagement and retention into a parent-centered, eco-developmental HIV preventive intervention for Hispanic adolescents and their families. *Journal of Pediatric Psychology*, 31(9), 874–890. <https://doi.org/10.1093/jpepsy/jsj046>.
- Raudenbush, S. W., & Liu, X. F. (2001). Effects of study duration, frequency of observation, and sample size on power in studies of group differences in polynomial change. *Psychological Methods*, 6(4), 387. <https://doi.org/10.1037/1082-989X.6.4.387>.
- Reaves, S., McMahon, S. D., Duffy, S. N., & Ruiz, L. (2018). The test of time: A meta-analytic review of the relation between school climate and problem behavior. *Aggression and Violent Behavior*, 39, 100–108. <https://doi.org/10.1016/j.avb.2018.01.006>.
- Redding, C. (2019). A teacher like me: A review of the effect of student-teacher racial/ethnic matching on teacher perceptions of students and student academic and behavioral outcomes. *Review of Educational Research*, 89(4), 499–535. <https://doi.org/10.3102/0034654319853545>.
- Reynolds, C. R., & Kamphaus, R. W. (2004). *Behavior assessment system for children, (BASC-2)* (vol. 5, pp. 33–44). Circle Pines, MN: American Guidance Service.
- Salbach-Andrae, H., Lenz, K., & Lehmkuhl, U. (2009). Patterns of agreement among parent, teacher and youth ratings in a referred sample. *European Psychiatry*, 24(5), 345–351. <https://doi.org/10.1016/j.eurpsy.2008.07.008>.
- Santiago, C. D., Raviv, T., & Jaycox, L. H. (2018). The rationale for school-based interventions. In C. D. In, T. R. Santiago, & L. H. Jaycox (Eds.), *Creating healing school communities: school-based interventions for students exposed to trauma* (pp. 27–35). American Psychological Association. <https://doi.org/10.1037/0000072-004>.
- Schochet, P. Z. (2009). An approach for addressing the multiple testing problem in social policy impact evaluations. *Evaluation Review*, 33(6), 539–567. <https://doi.org/10.1177/0193841X09350590>.
- Schuelke, M., Terry, R., & Day, E. (2013). Growth spline modeling. In *Paper 285–2013 in Proceedings of the 2003 SAS Global Forum*.
- Steeger, C. M., Buckley, P. R., Pampel, F. C., Gust, C. J., & Hill, K. G. (2021). Common methodological problems in randomized controlled trials of preventive interventions. *Prevention Science*, 22(8), 1159–1172. <https://doi.org/10.1007/s11121-021-01263-2>.
- Steinberg, L. (2016). Commentary on special issue on the adolescent brain: Redefining adolescence. *Neuroscience and Biobehavioral Reviews*, 70, 343–346. <https://doi.org/10.1016/j.neubiorev.2016.06.016>.
- Utrzan, D. S., Piehler, T. F., & Dishion, T. J. (2018). The role of deviant peers in Oppositional Defiant Disorder and Conduct Disorder. In J. E. Lochman, & W. Matthys (Eds.), *The Wiley handbook of Disruptive and Impulse-Control Disorders* (pp. 339–351). John Wiley & Sons. <https://doi.org/10.1002/9781119092254.ch21>.
- Wang, M.-T., & Eccles, J. S. (2012). Social support matters: Longitudinal effects of social support on three dimensions of school engagement from middle to high school. *Child Development*, 83(3), 877–895. <https://doi.org/10.1111/j.1467-8624.2012.01745.x>.
- Weist, P. S., Garbacz, A., Schultz, B., Bradshaw, C. P., & Lane, K. L. (2024). Revisiting the percentage of k-12 students in need of preventive interventions in schools in a “peri-COVID” era: Implications for the implementation of tiered programming. *Prevention Science*, 25(3), 481–487. <https://doi.org/10.1007/s11121-023-01618-x>.
- Whipp, A. M., Korhonen, T., Raevuori, A., Heikkilä, K., Pulkkinen, L., Rose, R. J., ... Vuoksima, E. (2019). Early adolescent aggression predicts antisocial personality disorder in young adults: A population-based study. *European Child and Adolescent Psychiatry*, 28, 341–350. <https://doi.org/10.1007/s00787-018-1198-9>.
- Wilson, D. B., Gottfredson, D. C., & Najaka, S. S. (2001). School-based prevention of problem behaviors: A met-analysis. *Journal of Quantitative Criminology*, 17(3), 247–272. <https://doi.org/10.1023/A:1011050217296>.
- Wymer, S. C., Corbin, C. M., & Williford, A. P. (2022). The relation between teacher and child race, teacher perceptions of disruptive behavior, and exclusionary discipline in preschool. *Journal of School Psychology*, 90, 33–42. <https://doi.org/10.1016/j.jsp.2021.10.003>.